Week 9: Dispersive Waves

- Wave motion is a central topic in many branches of physics.
- Waves transport energy (and momentum) without transporting mass whereas throwing a ball transports both energy and mass.
- This distinction is exact only for special limits (often for tiny waves) but these turn out to have many applications (e.g. sound).
- Often the theoretical description of waves is developed using so called “model equations”, but we will also develop the case of “deep water waves” on the surface of a fluid from first principles later in the course.
- The surface water wave derivation has many steps, and so for this slide deck I stick to model equations only.
- In an independent application slide deck, I use a model equation to discuss nonlinear waves and shocks.

The classical wave equation, and the one way advection equation

$$A_{tt} = c^2 A_{xx}$$

$$A(x,t) = F(x - ct) + G(x + ct)$$

D'Alembert's solution

rightward propagating waves

leftward propagating waves

- **Initial conditions determine wave shape, which does not change.**
- **Information propagates along special curves called characteristics**

$\frac{dx}{dt} = c$ **and** $\frac{dx}{dt} = -c$**.**

$$B_t = -cB_x$$

$$B(x,t) = F(\xi)$$

$$\xi = x - ct$$

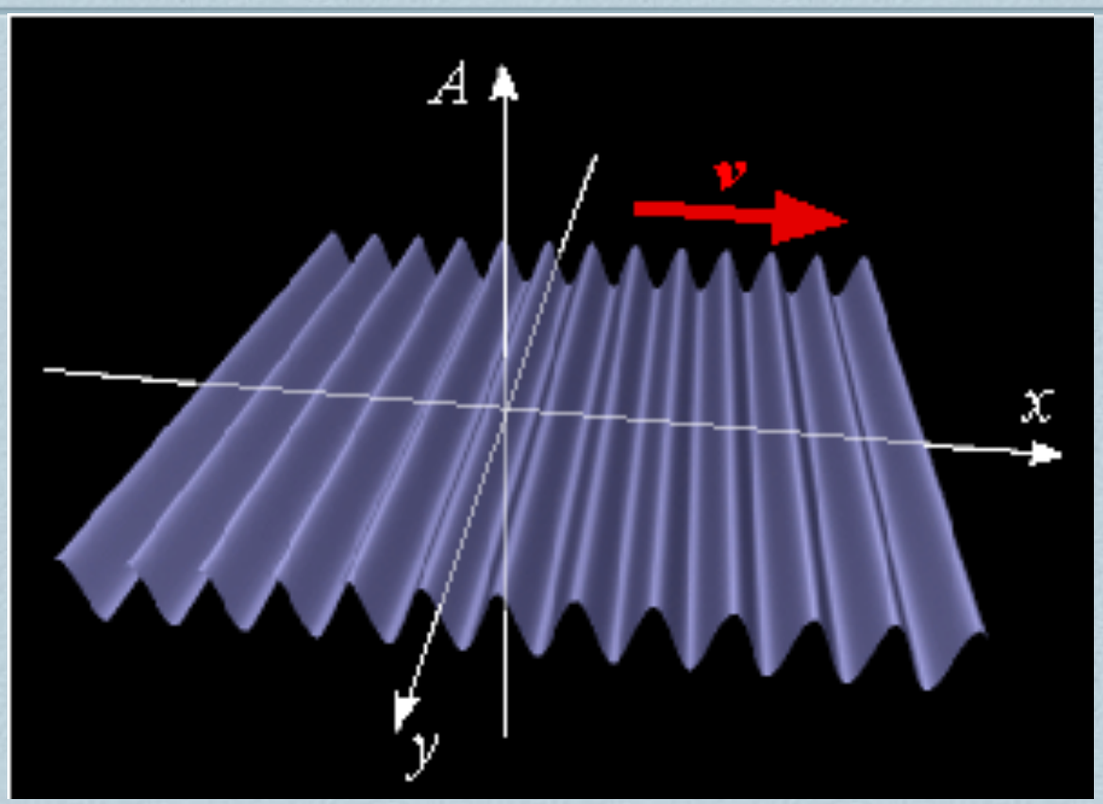
$$B_t = F'(\xi)\frac{\partial \xi}{\partial t} = F'(\xi)(-c)$$

$$B_x = F'(\xi)\frac{\partial \xi}{\partial x} = F'(\xi)$$

rightward propagating waves only

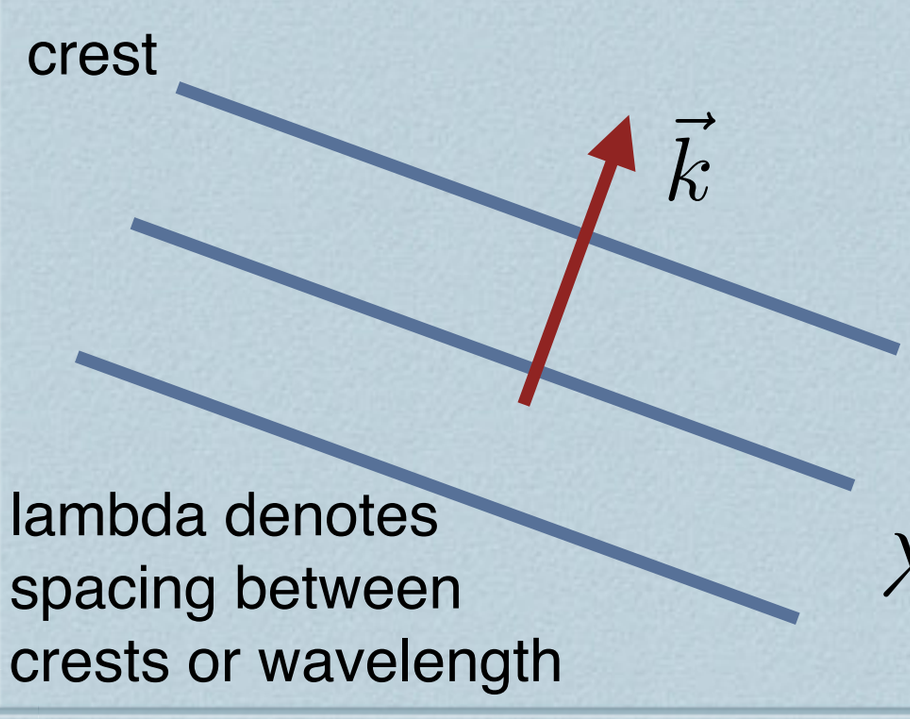
$$B(x,t) = F(x - ct)$$

D'Alembert's solution



$$A(x,t) = Re\{\exp[i(\vec{k}\cdot\vec{x} - \sigma t)]\}$$

- Plane waves are a basic building block of physical theories (next slide has examples).
- The direction of propagation and the inter-crest spacing is given by the wavenumber vector.
- In general the frequency may depend on wavenumber, though often the first examples we see (e.g. EM waves in a vacuum) are so-called non-dispersive waves.

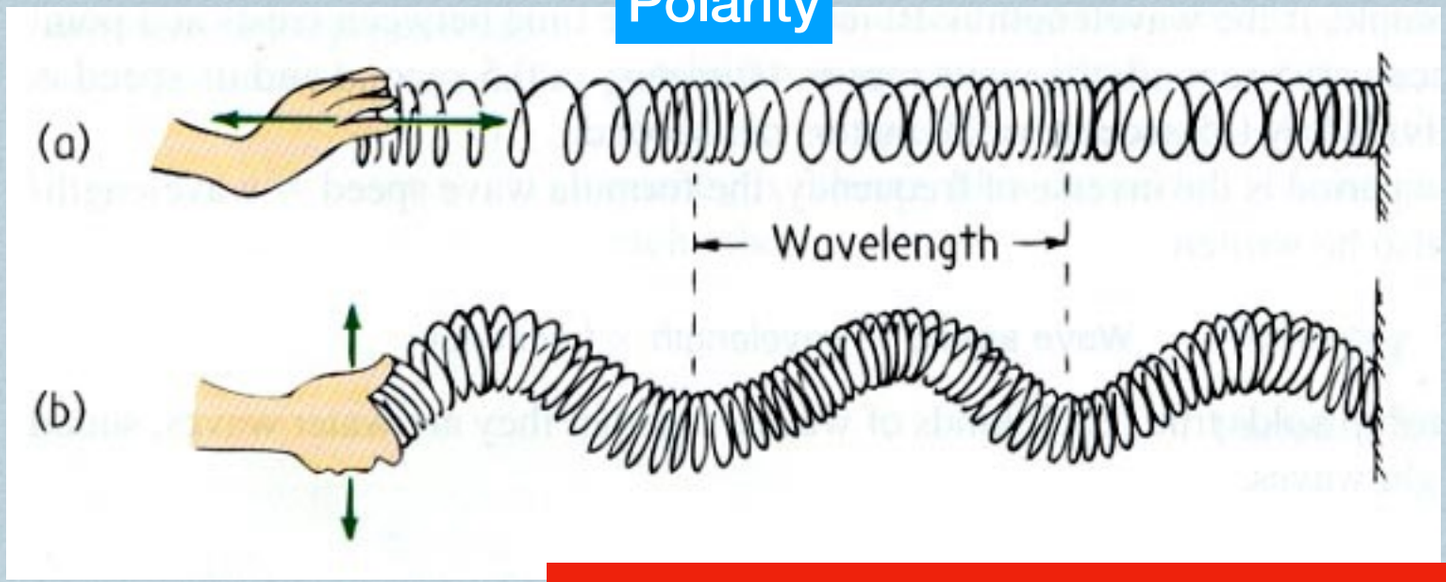


$$\lambda = \frac{2\pi}{|\vec{k}|}$$

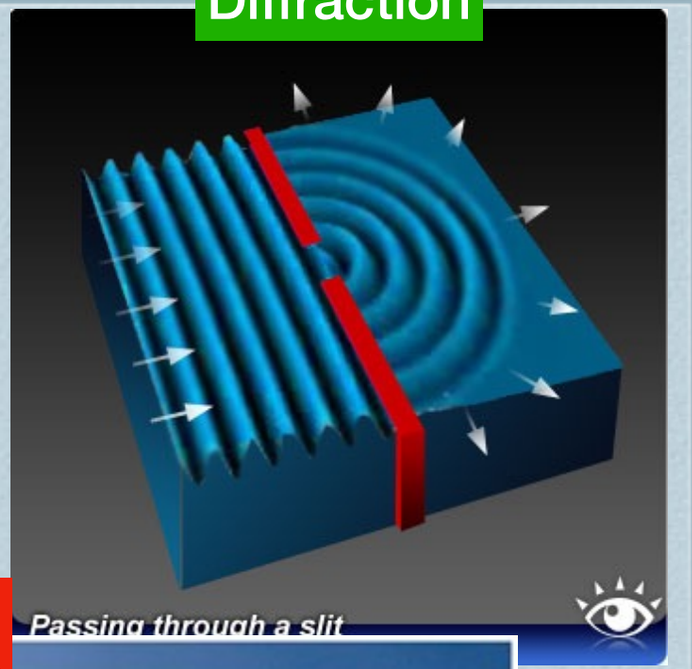
- Mathematicians are often presented "waves" as solutions to the wave equation (last slide) which is a bit unphysical.
- The wave equation is an example of a hyperbolic equation and such equations should really be thought of as representing signals (the information carried along characteristics), as opposed to waves.

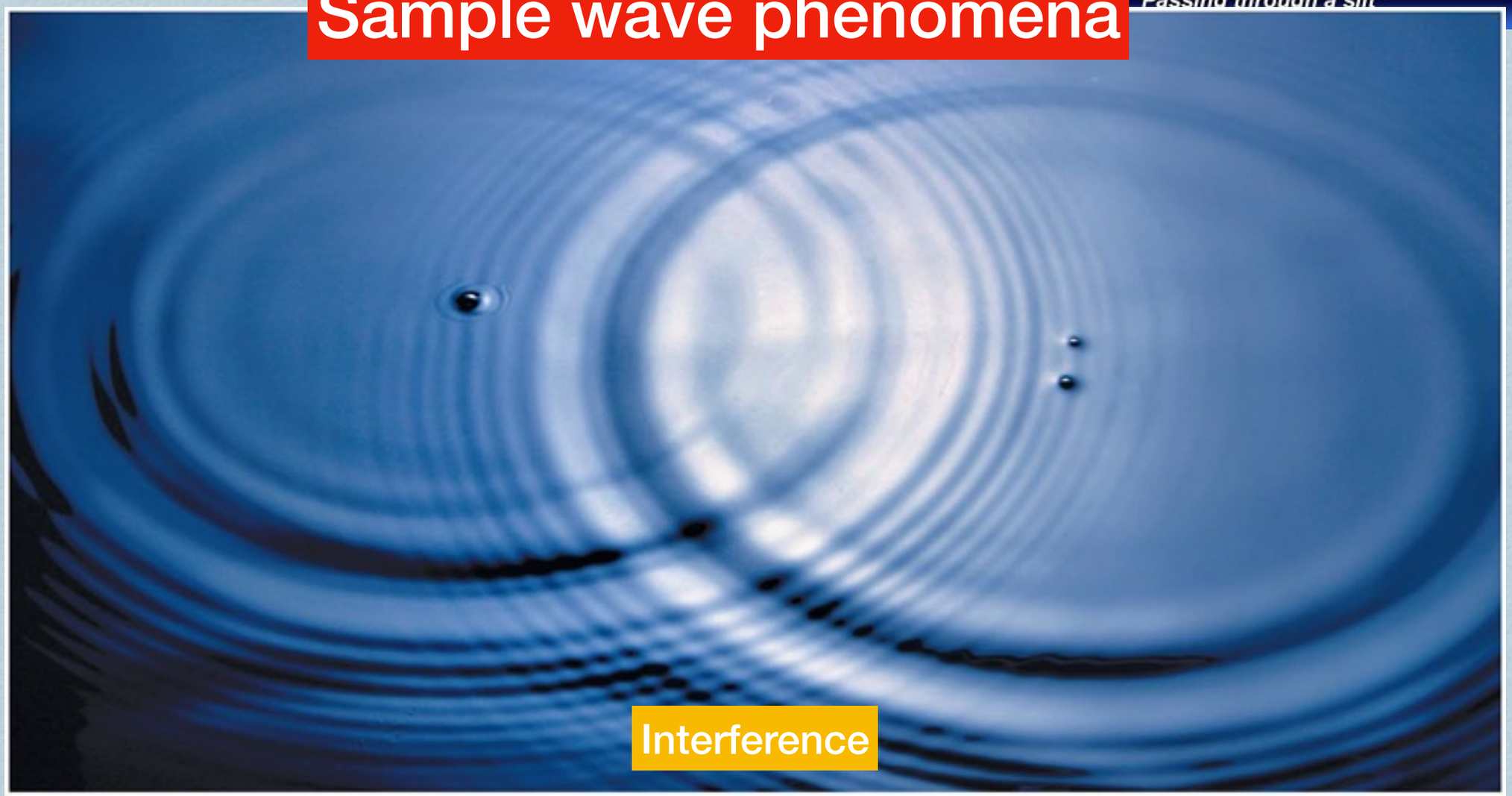
Sample wave phenomena

Polarity



Diffraction





Interference

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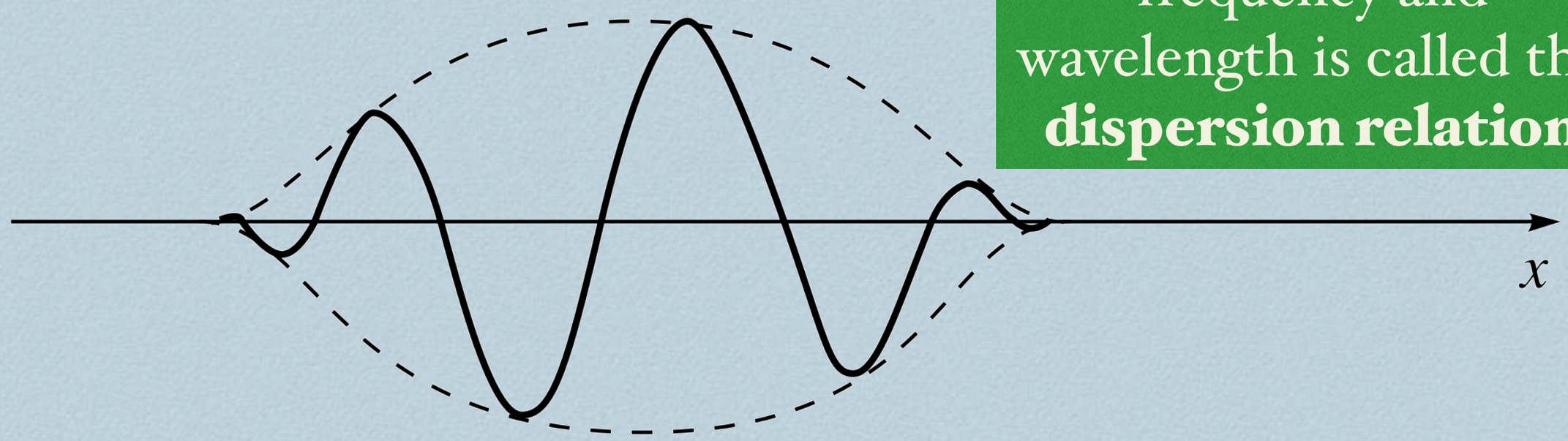
- Plane waves are quite idealized and often we are concerned with localized disturbances.

- These too have a well-known idealization in physics called a **wave packet**.

- A wave packet consists of a plane wave and an envelope. The envelope (dashed line below) is assumed to be much longer, and much more slowly varying than the plane wave (solid line below).

$$A(\epsilon x, t)\cos(kx - \sigma(k)t) \text{ where } \epsilon \ll 1$$

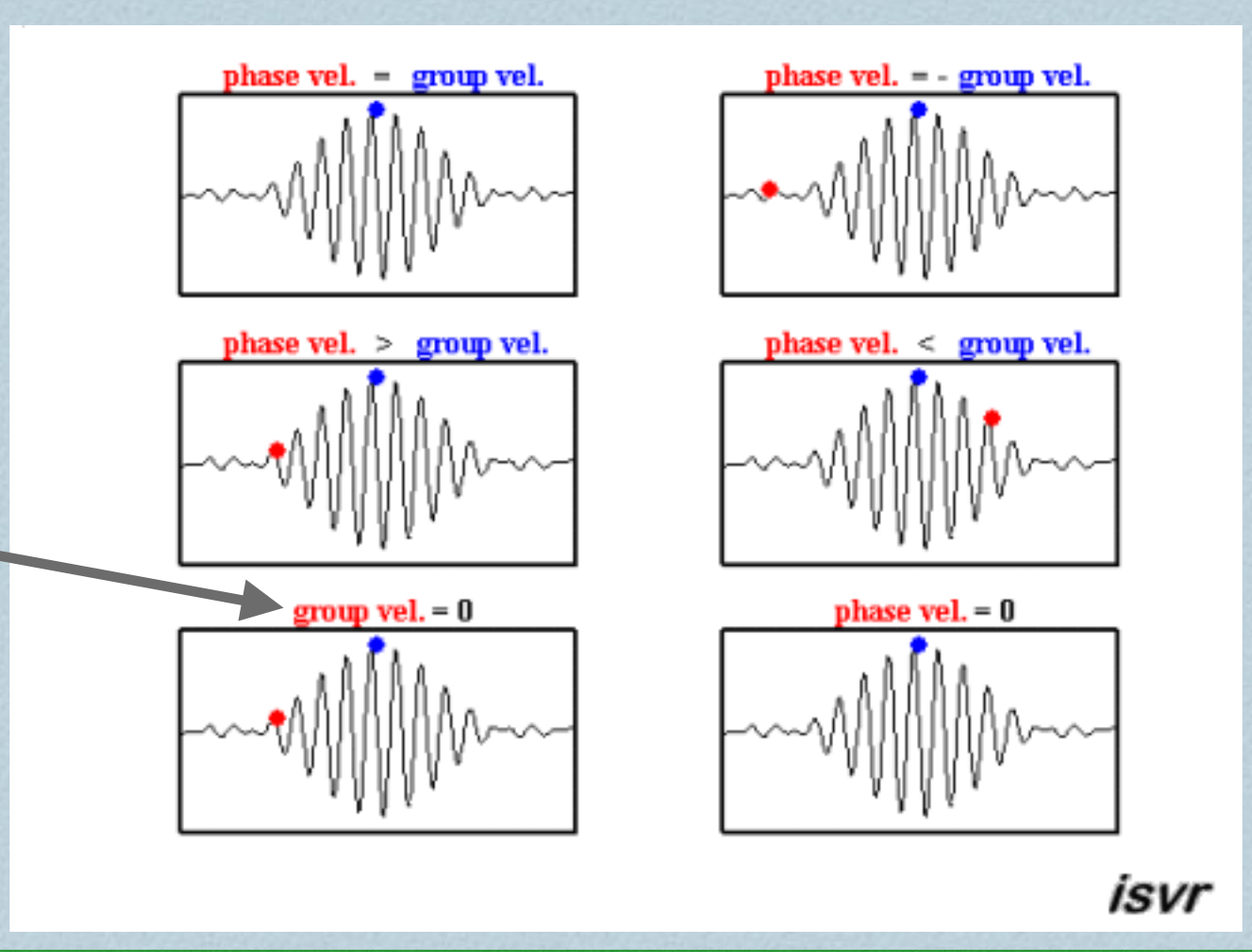
The relation between frequency and wavelength is called the **dispersion relation**



wave packet

- When frequency depends on wavenumber there are two velocities (or speeds).
- The velocity of the wave crest, called the **phase velocity**, which is largely a mathematical construct.
- The velocity of the wave packet, or of energy propagation, called the **group velocity** which is the more physical concept.

Group Velocity Should Be blue



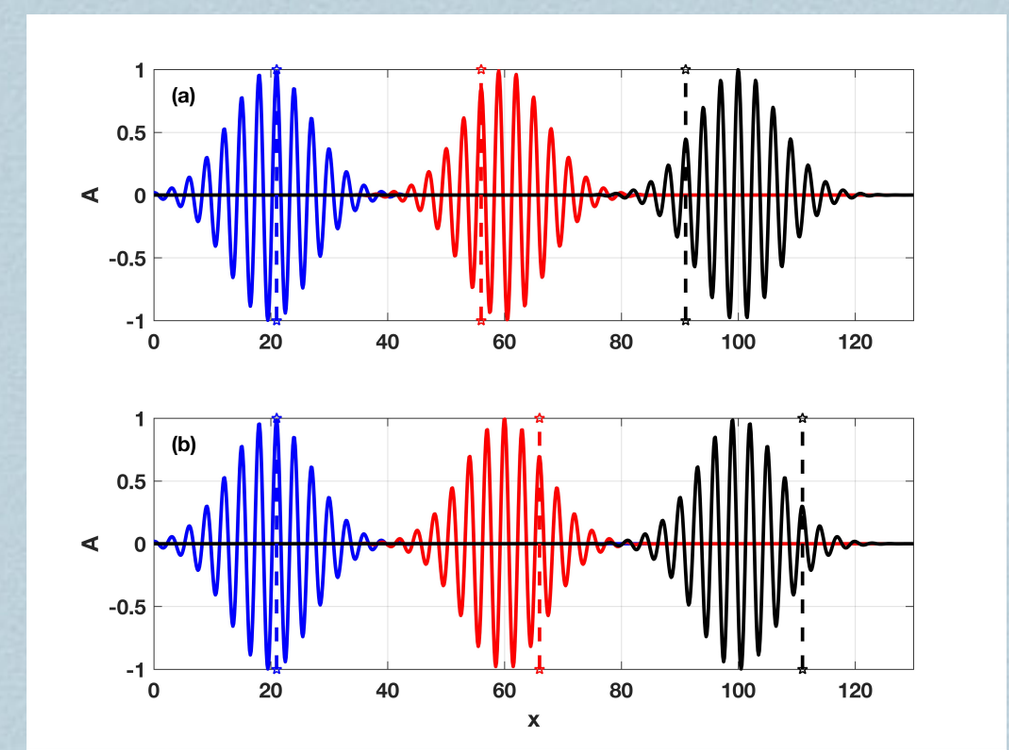
$$c_p = \frac{\sigma}{k}$$

$$c_g = \frac{d\sigma}{dk}$$

Examples of dispersion relations are shown in two slides

Same info as the last slide using snapshots as opposed to a movie

$c_p < c_g$



$c_p > c_g$

$$c_p = \frac{\sigma}{k}$$

The dashed vertical line moves with the phase velocity

$$c_g = \frac{d\sigma}{dk}$$

The packet moves with the group velocity

Examples of dispersion relations are shown on the next slide

Sample Dispersion Relations

$$\sigma = \pm\sqrt{gk} \qquad c_p = \pm\sqrt{\frac{g}{k}} \qquad c_g = \pm\frac{1}{2}\sqrt{\frac{g}{k}}$$

- Long waves travel faster, energy propagates slower than crests.
- Fairly common in nature, as we will see for deep water waves.

g is the acceleration due to gravity so a constant

$$\sigma = \pm\,\alpha k^{3/2} \qquad c_p = \pm\,\alpha\sqrt{k} \qquad c_g = \pm\,\alpha\frac{3}{2}\sqrt{k}$$

- Short waves travel faster, energy propagates faster than crests.
- Not common apart from surface tension (or capillary waves).

- Dispersion modified advection equation (a "model" equation)

$$A_t = -\,cA_x - \beta A_{xxx} \qquad A = \exp(i[kx - \sigma(k)t])$$

$$-i\sigma(k) = -\,cik + \beta ik^3 \qquad \sigma(k) = ck - \beta k^3$$

$$c_p = c - \beta k^2 \qquad c_g = c - 3\beta k^2 \leq c_p$$

Dispersive Waves

No restriction on equation form

$$A(x,t) = a_0 \exp[i(kx - \sigma(k)t)]$$

Frequency depends on wavelength

$$c_p = \frac{\sigma}{k}$$ phase speed: tracks crests

$$c_g = \frac{d\sigma}{dk}$$ group speed: tracks energy

If

$$A(x,0) = \sum_n a_n \cos(k_n x)$$

then

$$A(x,t) = \sum_n a_n \cos(k_n x - \sigma(k_n)t)$$

So waveform shape changes
But we have a recipe for how to reconstruct.

Hyperbolic Waves

Special equation form, e.g.

$$A_{tt} = c^2 A_{xx}$$ both directions

$$B_t = -cB_x$$ rightward only

Single propagation speed, c

D'Alembert solution:

$$B(x,t) = F(x - ct)$$

$$B(x,0) = F(x)$$

Initial conditions determine shape of wave which does not change with time.

Great Video of Dispersive Wave Applications
https://www.youtube.com/watch?v=95sQcSulRFM

What about these techniques for non-wave equations?

- **Consider the heat equation** $u_t = \nu u_{xx}$ **and try the wave ansatz** $u = \exp(i[kx - \sigma t])$
- **Substitution gives** $-i\sigma = -k^2\nu \implies \sigma = -i\nu k^2$
- **This means** $u = \exp(i[kx - \sigma t]) = \exp(ikx)\exp(-\nu k^2 t)$
- **Taking the real part we get** $u = \cos(kx)\exp(-\nu k^2 t)$
- **So instead of waves we get cosines damped in time**
- **The damping is faster as the wavenumber increases, or in other words short waves are damped faster**
- **This is one of the central properties of diffusion, and much easier to show using the wave ansatz.**
- **Try the exercise of fixing the frequency and solving for k. You should recover a solution like for the oscillating plate problem in viscous flow (different independent variable).**